# Souha Alioua

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## PROFESSIONAL SUMMARY

Business intelligence and analytics professional who builds decision tools (self-service dashboards, automated data pipelines, and statistical models) using SQL, Python, APIs, and BI platforms (Tableau, Power BI, Looker Studio). Skilled in translating stakeholder requirements into scalable reporting solutions and turning complex demographic and economic data into insights that drive planning and operational decisions.

## TECHNICAL SKILLS

- **Programming & Data:** Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, JavaScript, HTML/CSS
- **Databases & ETL:** SQL Server, MySQL, MS Access, APIs, data cleaning, ETL, data validation
- **BI & Visualization:** Tableau, Power BI, Looker Studio, IBM Cognos Analytics, ArcGIS
- **Tools:** Git, GitLab, Excel, PowerPoint
- **Languages:** English, French, Arabic
- **Certification:** IBM Data Analytics Professional Certificate

## PROFESSIONAL EXPERIENCE

**Maryland Department of Planning – State Data & Geospatial Analysis Center** Baltimore, MD

*Data Analyst* *Feb 2025 – Present*

- Delivered an interactive Economic Census application end-to-end (scoping, build, and deployment) giving planners and analysts self-service comparisons of employment, payroll, revenue, and establishment trends across Maryland industries and counties; built with Python, Census APIs, SQL Server, and JavaScript. View project
- Built a public-facing economic dashboard integrating BEA data on GDP, personal income, industry earnings, and consumer spending, with interactive geographic comparisons and inflation-adjusted measures (in development).
- Engineered 10+ automated data pipelines (Python, APIs) for demographic, economic, and health datasets, eliminating ~10 hours of manual reporting per week and ensuring consistent, on-time delivery to downstream teams.
- Designed and maintained SQL Server datasets for recurring reporting, implementing validation, reconciliation, and documentation procedures to improve data accuracy and reproducibility.
- Produced official population projections for 157 Maryland municipalities used in state planning, benchmarking statistical and machine-learning methods to ensure accuracy while maintaining consistency with county-level totals.

## SELECTED PROJECTS

**Small-Area Population Projection Research | Independent Research** 2025 – Present

- Developed a statistical model for municipal population projections that adjusts estimates based on population size and historical stability while maintaining consistency with county totals.
- Compared traditional statistical and machine-learning methods through historical backtesting across multiple forecasting periods.
- Evaluated performance using MAE, MAPE, RMSE, median percentage error, and bias to identify accuracy tradeoffs across municipalities.

**Educational Attainment & Well-Being in Maryland |** View project 2026

- Integrated ACS, BLS, County Health Rankings, and TIGER/Line data to analyze educational attainment and well-being across Maryland counties and census tracts.
- Conducted county-level regression and outlier analysis across economic, health, behavioral, and housing indicators.
- Built an accessible interactive report with JavaScript, HTML/CSS, choropleth maps, scatterplots, and documented data sources and methodology.

## EDUCATION

**Dickinson College** Carlisle, PA

*B.S. in Mathematics* *Dec 2024*

- Honors: Henry P. Cannon Memorial Prize in Mathematics; John Dickinson Scholarship; Dean's List